

Danh mục: medication

Type 2 diabetes in adults: management

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NICE guideline

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NG28

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Download guidance (PDF)

Reviewing medicines

Recommendations in this section that cover dipeptidyl peptidase-4 (DPP-4) inhibitors, glucagonlike peptide-1 (GLP-1) receptor agonists, sulfonylureas and sodium–glucose cotransporter-2 (SGLT-2) inhibitors refer to each of these groups of medicines at class level.

For GLP-1 receptor agonists, at the time of publication (February 2026) this only includes liraglutide, dulaglutide, and semaglutide. For subcutaneous semaglutide (Ozempic), this only includes doses up to 1 mg once a week.

1.22 Principles

1.22.1

When reviewing treatments, make a shared decision about changes with the person with type 2 diabetes. See the recommendations on involving people in medicine discussions

[2022, amended 2026]

1.22.2

Optimise their current treatment regimen before changing treatments, taking into account factors such as:

adverse effects

prescribed doses and formulations

adherence to, and management of existing medicines

the need to revisit advice about diet and healthy living.

See also the

recommendations on individualised care

[2022, amended 2026]

1.22.3

If response to medicines suggests that type 2 diabetes might not be the correct diagnosis, see the recommendations on initial diagnosis and revisiting initial diagnosis in NICE's guideline on managing type 1 diabetes [2026]

1.23 Reviewing metformin

1.23.1

For adults with type 2 diabetes who are already taking standard-release metformin:

continue with this treatment or

switch to modified-release metformin if standard-release metformin is not tolerated or if this is the person's preference.

[2026]

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Why the committee made this recommendation

The committee agreed that, when changes to treatment are being considered, it is important to review existing treatment options first. Stopping medications that have not worked, for example, in controlling blood glucose, and optimising current treatments may remove the need to prescribe additional medicines. In particular, there might be factors, such as problems with adherence or adverse effects, that might make existing treatments less effective or ineffective. Addressing these might mean that adding a new medicine is unnecessary.

In the committee's experience, some people have their medications stopped after they reach their glycaemic targets. This can lead to their HbA1c levels and weight rising again. Often, it would be better for the person to keep taking medications that have helped them reach their individualised glycaemic targets, to prevent future problems. There was no evidence on which groups would most benefit from this, so the decision would need to be based on clinical judgement and the preferences of the person with type 2 diabetes.

SGLT-2 inhibitors are a good treatment option for most people and provide cardiovascular and renal protection that cannot be measured by tests. Therefore, the committee agreed that these should be continued even if they do not help the person reach their individualised glycaemic targets.

However, the committee acknowledged that the decision is more complicated for GLP-1 receptor agonists and tirzepatide. For people with atherosclerotic cardiovascular disease or early onset type 2 diabetes, GLP-1 receptor agonists and SGLT-2 inhibitors are being used to prevent cardiovascular events. For these groups, continuing GLP-1 receptor agonists can provide benefits even if they do not help the person reach their glycaemic targets. For people who do not have atherosclerotic cardiovascular disease or early onset type 2 diabetes, GLP-1 receptor agonists and tirzepatide are being used to reach individualised glycaemic targets, so they should be treated like any other medication and stopped if they are not effective for this purpose.

The list of factors to think about as part of optimisation is not exhaustive but includes those that the committee thought were particularly important. The committee agreed that it is important to revisit advice about diet and healthy living. This is because part of this discussion is to ensure the person is supported with both non-pharmacological and pharmacological interventions to improve their current health and prognosis.

The committee agreed that there are cases where treatment with certain medicines highlights diagnostic uncertainty (for example: absence of response to treatments other than insulin, sudden unexpected weight loss). Therefore, the committee highlighted guidance regarding type 1 diabetes and revisiting other diagnoses.

People already on standard-release metformin

There was no evidence identified in the review to show that modified-release metformin was more effective than standard-release metformin, and no evidence that it would be more cost effective for people for whom it works. However, the committee agreed that, in their clinical experience, people can experience fewer gastrointestinal adverse events with modified-release metformin compared to standard-release metformin. Additionally, a person may want to reduce the number of times they take metformin each day. Therefore, the option of switching from standard-release to modified-release metformin should be available.

Not combining a DPP-4 inhibitor and a GLP-1 agonist

Based on their own experience, the committee agreed that combining a GLP-1 receptor agonist or tirzepatide and a DPP-4 inhibitor would not add value. The 2 medicines have a similar mechanism of action, as they act on different parts of the GLP-1 pathway. Because of this, it is unlikely that combining the medications provides any additional effect and so it is unlikely to be clinically or cost effective.

Full details of the evidence and the committee's discussion are in:

evidence review E: initial management

Danh mục: medication

evidence review F: subsequent management

How the recommendation might affect practice

The recommendations will lead to people taking SGLT-2 inhibitors and GLP-1 receptor agonists or tirzepatide for longer. This will initially increase costs. However, the long-term protective benefits of these medicines will reduce the need to treat future cardiovascular and renal problems, which will lead to cost savings. Otherwise, the recommendations are not expected to change current practice significantly.

The 2022 recommendations about reviewing medicines are not expected to be a change in practice or to need substantial additional resources because these conversations should already take place.

1.24 Reviewing other medicines

1.24.1

If the person has reached their individualised

glycaemic target

and weight target (as defined in the

section on discussing results and referral in NICE's guideline on overweight and obesity

), consider continuing any medicines that have contributed to this.

[2026]

1.24.2

Consider continuing SGLT-2 inhibitors for their cardiovascular or renal benefits, even if they do not help the person reach their individualised glycaemic targets.

[2026]

1.24.3

Stop GLP-1 receptor agonists or tirzepatide if the person becomes underweight (BMI under 18.5 kg/m

[2026]

1.24.4

Stop GLP-1 receptor agonists or tirzepatide if they do not help the person reach their individualised glycaemic targets and they are not being taken for their cardiovascular benefits.

[2026]

1.24.5

Take into account adverse effects from combining medicines (for example hypoglycaemia).

[2022, amended 2026]

1.24.6

Do not offer both a GLP-1 receptor agonist or tirzepatide and a DPP-4 inhibitor together to treat type 2 diabetes.

[2026]

Why the committee made these recommendations

The committee agreed that, when changes to treatment are being considered, it is important to review existing treatment options first. Stopping medications that have not worked, for example, in controlling blood glucose, and optimising current treatments may remove the need to prescribe additional medicines. In particular, there might be factors, such as problems with adherence or adverse effects, that might make existing treatments less effective or ineffective. Addressing these might mean that adding a new medicine is unnecessary.

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